

Ransomware Detection System and Analysis Using Latest Tool

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Abstract: *Today ransomware has a serious threat to the online world. Most of the software firms, universities, companies, and organizations in the world are trying to take alert decisions to save from ransomware attacks. A joint statement was issued by both governments of the United States and about ransomware attacks insistence users to stay alert and take precautions. Recently on May 19th, 2017, the Swiss government observed the Ransomware Info Day, to spread awareness regarding ransomware and its effects. Ransomware in India as well is on the rise. Ransomware is a combination of two words, ransomware and software. In other words, ransomware makes it really hard for users to access all files and data in a computer because it hacks the data and do not allow the users to access the files unless they pay ransom to the hacker. Dr. Joseph Popp is the first person to develop ransomware. Ransomware is known as a global epidemic that attacks all types of organizations using computing infrastructure. Ransomware has undergone various evolutions. Preventive measures are needed in order to avoid losses especially in business. The purpose of this study is to examine the evolution of ransomware and preventions against it.*

Keywords: Ransomware, Encryption-Decryption, Wannacry, Tools.

I. INTRODUCTION

A Ransomware is a type of malware that locks your files, data or the pc itself and extorts money from you to provide access. This can be another way for malware developers to 'collect funds' for their ill-conceived exercises on the web. Once ransomware contaminates a network device, it begins scrambling records, organizers, and whole hard drives partitions utilizing encryption algorithms like RSA or RC4.

The development, distribution and successful attacks of ransomware has grown exponentially over the past 4 years. According to Trend Micro research [1], 2016 saw a record 400% rise in new ransomware families (roughly 150 new families). It is thus evident that major market share anti-virus solutions are failing to contain the threat of ransomware. The inadequacies of current solutions lie their heavy reliance on static-based detection techniques.

The ransomware is partitioned into two types: Lock Screen Ransomware and Encryption Ransomware Lock screen ransomware locks system and demands a ransom for letting you access it once again. The second type, i.e. the Encryption ransomware, modify the files in your system and demands a ransom to decrypt them again. The other types of ransomware are: 1. Master Boot Record (MBR) ransomware. 2. Ransomware encrypting web servers. 3. Android mobile device ransomware. 4. IoT Ransomware

II. LITERACY SURVEY

In this section detailed overview of the current site of ransomware attacks posed.

2.1 Identify, Research and Collect Idea

Advanced ransomware like Spore, WannaCrypt (moreover known as WannaCry). And Petya (moreover known as NotPetya) spread to other computers through network shares or exploits. Spora drops ransomware duplicates in

organizing shares. 2. WannaCrypt abuses the Server Message Block (SMB) Powerlessness CVE-2017-0144 (too called Eternal Blue) to contaminate other computers [10].

2.2 Ransomware Attacks On Append Year

to encrypt files and the command and control methods. [1]

Families	Deployment	Date Appeared	Cryptographic Techniques	Command and Control
Reveton	Drive-by downloads	2012	RSA and DES	MoneyPak
GpCode	Email Attachments	2013	660-bit RSA and AES	Tor Network
CryptoLocker	Compromised websites and Email Attachments	2013	2048-bit RSA	Tor Network
CryptoWall	Compromised websites and Email Attachments	2013	2048-bit RSA	Tor Network
FileCrypto	Compromised websites and Email Attachments	2013	2048-bit RSA	Tor Network
TeslaCrypt	Compromised websites and Email Attachments	2013	2048-bit RSA	Tor Network
CTB-Locker	Email Attachments	2014	ECC	Onion Network
Shade	Spam Email	2015	RSA-3072 and AES-256	Fixed Server
Jigsaw	Word Document with Javascript	2016	RSA and AES	Onion Network
WannaCry	Samba Vulnerability	2017	RSA and AES	Onion Network

Ransommware Attacks on

- System – Integrated network
- Attacks on Android

2.3 Overview

Ransomware are now delivered as Word macros and PowerShell scripts. ‘Petya’ encrypted hard drive master boot record (MBR), as well as files, rendering computers completely unusable. The MBR is replaced with the malware’s own bootloader so that the ransom note can be displayed. The most common method of delivering ransomware is the phishing attack and it is not easily recoverable.

According to the Federal Bureau of Investigation (FBI), estimated losses of about one billion US dollars (\$1 billion) was incurred to ransomware attacks in the year 2016. The boom recorded by this crime shows that a good number of victims eventually pay the ransom to have their data unlocked. Nearly 40 percent of ransomware victims paid the ransom. Three out of four ransomware gangs are willing to negotiate prices for decryption. On average, they will give a 29% discount on the fee initially demanded. Unfortunately, traditional preventive and reactive security measures are not adequate to handle the effect of ransomware attacks [6].

In this paper, we provide a brief overview of the current trend, challenges, and research progress in the bid to finding lasting solutions to the menace of ransomware that currently challenge computer and network security, and data privacy.

III. FAILED RANSOMWARE ATTACKS

3.1 Fake Windows 10 Lock Screen

It tells the user that their license has expired, turns out to have the decryption key buried in the code. Researchers from Symantec discovered that, while the criminals had gone to considerable effort to set up fake tech support websites for the scam, the phone number they gave out for victims to call was never answered and was soon disconnected. On reverse engineering the code, the researchers found the decryption key (8716098676542789) plainly visible.

IV. HITLER RANSOMWARE

It claims to have encrypted the victim's files, but in fact simply deletes file extensions for anything found in certain directories. After an hour it crashes the PC and, on reboot, deletes the files. The payment demanded is a cash code for E25 Euro Vodafone Card. Text found in the code suggests it originates in Germany.

4.1 Major Ransomware Attacks

A. Encrypting Ransomware

This class of malware, once executed, silently searches for and encrypts valuable files on the victim's machine. After the first step is completed, a message is displayed to the user that asks for a ransom in return of the hidden files. Detailed instructions are presented to the user, sometimes, even a call center number is provided. After the ransom is paid, the victim will be given either a key or code for the decryption of the files, or an executable file that is created specifically to decrypt the files on that victim's machine. Examples of typical crypto ransomware include CryptoWall, CryptoLocker, WannaCry and Locky.

B. Non-encrypting Ransomware

Typically, these types of ransomware use means of locking or locking access to the target machine and demanding a ransom or asking for a user action that ends up costing money to unlock. To make the users pay the ransom, some of these malware either ask their victims for payment up front, while others rely on deceit (ex. Ask the user to call a high rate phone number and presenting the call as free.). Examples of typical locker-ransomware include Winlocker and Reveton.

C. Leakware

(also called Doxware) This class of malware differ from the ones presented above in that they don't block access to the victim's machine or any information that it is stored on it. Rather, it silently collects sensitive information from the machine and uses it to blackmail the victim. The information gathered is stored on servers or other infected machines and the attacker threatens the victim that the data will be published if payment is not made. 1.4. Mobile ransomware These ransomware target mobile devices (phones, tablets, etc). Because, usually, the information is easy to back up and restore to and from the cloud or any back-up solution, these malware act as blockers, with little incentive to encrypt data. Thus, unlike regular locker malware, they rely on the value of the mobile device rather for the ransom, rather than on the information that is saved on it. Once these malware are installed or ran on the victim machine, some attempt to display a blocking message on top of the UI, while others use a form of click high-jacking to cause the user to allow it higher privileges.

4.2 Ransomware Behavior

Ransomware have been classified as a malware subcategory. As such, they have a number of common traits to other malware. These traits can be organized under six categories:

A. Payload Persistence

The attack is carried out all the way; usually done through placing an executable in the startup folder, scheduling a task on startup or registering a startup item in the registry

B. Anti-system Restore

Prevent system restore from rolling back the changes; usually done by deleting shadow copy saves.

C. Stealth Techniques

Malware will try to run in a stealthy manner to avoid detection; usually done by injecting code into legitimate processes, running from AppData or naming executables after other Windows processes .

D. Environment Mapping

Some ransomware analyze the system they infected; this is done to determine the value of the target, other possible targets (over the network) and if it's running on a real computer or on a sandbox environment that could be attempting to analyze it.

E. Network Traffic

Most ransomware require an internet connection; they use it for downloading the payload related files and/or for the communication of the encryption key.

F. Privilege Elevation

Using elevated privileges carry out the attack; sometimes, simply asking for administrator access may work or other privilege escalation techniques may be used, like click high-jacking.

V. METHODOLOGY

Tools and strategies for analysing ransomware

The development and use of sandboxes in the security industry has enabled a secure environment for the activation and analysis of malicious samples. Monitoring tools are integrated into sandboxes to observe and report on the sample's behaviour at the user and kernel-level. Malware analysis is available online at VirusTotal.com, hybrid-analysis.com and Malwr.com, as a bare-metal sandbox such as Barecloud and BareBox (Yokoyama et al. 2016), and as a package such as RanSim (KnowBe4 2017), REMnux (Zeltser 2014), Cisco (Umbrella 2016; Zscaler 2016; SonicWall 2016) and the well-known Cuckoo Sandbox (Ferrand 2015; Yokoyama et al. 2016; Kharraz et al. 2016). Cuckoo Sandbox allows the submission of Dynamic Linked Libraries (DDLs), Java files, binary executables, URLs, MS Office documents, and PDFs as samples (Ferrand 2015).

VI. RESULTS, ANALYSIS AND DISCUSSION

This section presents the results and analysis of applying the Randep model on 18 families of ransomware, along with the results and analysis of the user study. Each part is accompanied by relevant discussion to explain the findings and insights gained from the research.

VII. MODEL OF PREDICTIVE NATURE OF RANSOMWARE

If we look at the higher level, ransomware (in particular, crypto-ransomware) will likely have three stages: *stealth* (in which its main priority is to remain undetected while it prepares the groundwork for the ransomware attack), *suspicious* (in which it starts carrying out the damaging part of the attack, but it may not be detected straight away), and *obvious* (in which it makes its presence known to its victim, namely by notifying of its demand through a threatening message, and by deleting the victim's files). The transition at the higher level is pretty straightforward: stealth, followed by suspicious and then finally obvious.

Looking deeper, there are several lower level stages that ransomware may exhibit. These are probabilistic in nature, in a sense that not all ransomware strains will have all of these stages and/or the transition sequence between stages may differ. The lower level stages are:

- Fingerprint creating signatures of the OS's features and determining suitability for payload deployment.
- Propagate exploring the possibility of lateral movement within a network or connected devices.
- Communicate sending and receiving data from the attacker's C&C server.
- Map reading the contents of suitable files in the victim's environment.
- Encrypt encrypting potentially valuable data on the victim's computer.
- Lock reducing or disabling the availability of the OS to the victim.
- Delete overwriting or unlinking the contents of the victim's data.
- Threaten presenting a threatening message to force the victim to pay up.

Figure depicts our Randep predictive deployment model of ransomware. We have also developed a Randep classifier, which maps the Window API function calls, signatures, registration keys, and network calls into categories of the eight stages outlined above.

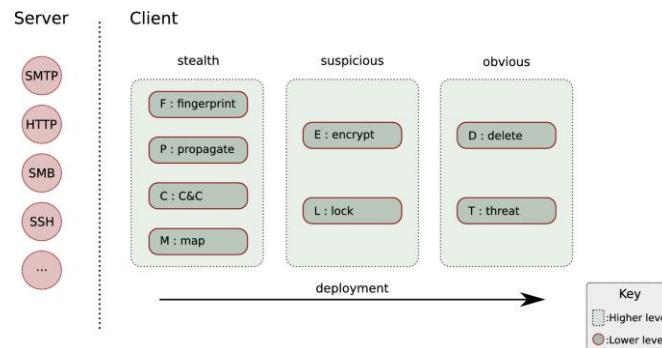


Figure 2

Predictive model of ransomware deployment methods

Lock-type ransomware would at least employ lock and threat stages. The majority of new ransomware families (> 95% in 2016) are of the crypto variety, therefore it is worth to focus on the actions of this type of ransomware.

7.1 Largest Ransomware Attack

The WannaCry ransomware attack was a worldwide cyberattack in May 2017 by the WannaCry ransomware cryptoworm, which targeted computers running the Microsoft Windows operating system by encrypting data and demanding ransom payments in the Bitcoin cryptocurrency.

7.2 WannaCry

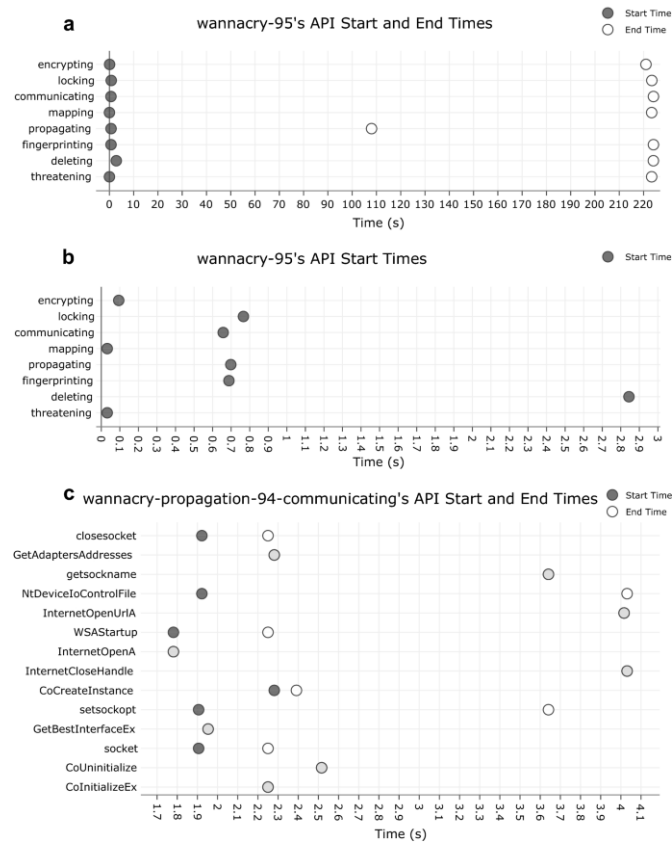
Two samples of WannaCry were analysed. The main signatures to identify the malware's maliciousness include its ability to unpack itself, anti-sandbox strategies, fingerprinting, manipulation of files and folders, and setup of the TOR router. Over 500 files were encrypted, the desktop background was changed to the message of the ransom, and a graphical user interface popped-up in the foreground of the user's screen.

Another variant of WannaCry, called mssecsvc.exe was also analysed. It carries out checks on the kill-switch domain name, and scans for open RDP connections. The sandbox was setup without modifying the hosts file to make the HTTP GET request to the kill-switch timeout, and without any open RDP connections. The sample scored 3.6 out of 10, and carried out four DNS lookups on: www.iuqerfsodp9ifjaposdfjhgosurijfaewrgwea.com which is the domain name used for the kill-switch. Since the address is still registered, the sample died. The communication state creates a server and binds it to 127.0.0.1 after 87 s, which WannaCry uses to send and receive packets over the TOR network. The malware uses TOR in an attempt to anonymize its network data, and to avoid detection. At 106.59 s, the malware makes a call to LookupPrivilegeValueW, which gets the privilege value and name of the logged-on user's locally unique identifier (LUID). In the propagation state we see the use of OpenSCManager after 107 s, which opens a connection and the service control manager database on a given computer. Then after 17 s the local server is shutdown.

Fig. 8

Randep states of WannaCry ransomware, showing a full-view, b zoomed-in, and c WannaCry's mssecsvc.exe process analysis showing communicating functions

WannaCry starts encryption early with a call to SetFileTime, it then sets up a new handle for the Cryptographic API functions, and decrypts a 16-byte string. The encryption of files begins at 2.84 s with a call to CryptGenKey, CryptExportKey and CryptEncrypt (CryptEncrypt carries out the encryption of the files from 2.84 to 60.83 s. The encrypted contents are temporarily stored in the system's default temporary folder, and the encrypted files replace the originals with a call to MoveFileWithProgressW at 3.68 s. The encryption ends when the original file has been replaced, which is noted by the end of MoveFileWithProgressW at 143.88 s. Hence the 3129 files encrypted took around 141 s, i.e. 22 files/s.



Tools-

Top 4 Best Ransomware Protection Programs

A new security solution or product is introduced in the cyber security market every day. Selecting the right tool catered to your enterprise security needs is critical. Let's look at the top 4 software programs available on the market today with the best ransomware protection and prevention capabilities.

1. ManageEngine OpManager
2. SolarWinds Patch Manager (PM)
3. Bitdefender Anti-Ransomware

Trend Micro Ransomware File Decryptor.

Some Fundamentals of ManageEngine OP Manager for Defending And Performing on Attacks

Fundamentals of ManageEngine are

DataSecurity Plus is a real-time change monitoring and alerting tool for Windows file systems. Since it uses dedicated agents to monitor files continuously, DataSecurity Plus has the ability to detect file changes the very instant they happen. This tool offers two important features which play a critical role in detecting and responding to ransomware attacks successfully: mass access alerts and automatic alert responses. Using these two features, DataSecurity Plus significantly reduces the time it takes to detect and respond to a ransomware attack. In fact, it automatically responds as soon as it detects the signs of a ransomware-type compromise. In doing so, it completely removes the need for human intervention, which is often slow and unsuccessful when pitted against ransomware attacks.

Mass Access Alerts

When an encryption attack is underway, the ransomware accesses and modifies an unusually large number of files in a short period of time. DataSecurity Plus can be configured to monitor the frequency of file modifications by a user, and to alert whenever the number of modifications crosses a specified threshold within a specified time period.

Automatic Alert Response

DataSecurity Plus allows you to configure a predetermined response to an alert. In other words, you can program the tool to take a specified action when a certain alert is triggered, effectively enabling you to automate the incident response. DataSecurity Plus has a built-in ransomware alert response, which locks down the infected device, thereby stopping the spread of ransomware to network storage or other systems and preventing the attacker from causing any further damage.

Working of ManageEngine OP Manager on Detection And Ransomware Attack

ManageEngine Desktop Central helps you to protect your computers from WannaCrypt ransomware. After detecting the computers that are vulnerable, Desktop Central lets you identify the computers that are missing critical patches and then deploy those patches immediately. You can also use Desktop Central's firewall configuration to block the vulnerable ports so that you can prevent WannaCrypt from spreading across your network.

WannaCrypt ransomware is a worm that targets computers running Windows operating systems that are not up-to-date.

For That Needs To Check On Patches-

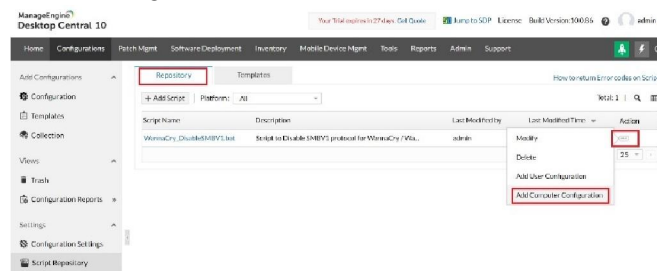
Patch ID 22058	Security Update for Windows Vista (KB4012598)
Patch ID 22059	Security Update for Windows Vista for x64-based Systems (KB4012598)
Patch ID 22060	Security Update for Windows Server 2008 (KB4012598)
Patch ID 22061	Security Update for Windows Server 2008 for x64-based Systems (KB4012598)
Patch ID 22513	Security Update for Windows XP SP3 (KB4012598)
Patch ID 22516	Security Update for Windows XP SP2 for x64-based Systems (KB4012598)
Patch ID 22514	Security Update for Windows Server 2003 (KB4012598)
Patch ID 22517	Security Update for Windows Server 2003 for x64-based Systems (KB4012598)
Patch ID 22515	Security Update for Windows 8 (KB4012598)

There are Many More Can Get On Internet.

Secure the network from WannaCrypt using Desktop Central

1. Deploying the critical patches to respective computers
2. Ensure that you Patch Vulnerability database is up to date. To update

Add the script, "WannaCry_DisableSMBV1.bat", to the repository. From the repository, deploy this script as a computer configuration to the desired target machines.



Blocks Critical Ports. Firewall Port go ON under Action on Firewall and select Block under Action on Ports. Select the port 137 which needed to be blocked. Block the other ports by clicking on Add More Ports.

It get protected from Ransomware Attack.

VIII. CONCLUSION & FUTURE SCOPE

In this research, the goal to improve malware detection, prevention, and mitigation. Using the method to reduce damage caused by ransomware attacks and techniques to minimize the ransomware attack loopholes in the network. In the Future, It will help to find malware more effective way with upcoming new technology and easy to minimize organization losses due to ransomware. Also, this report motivates new researchers and analytics for the decryption of infected files. Best Practice and Preventions of Ransomware Attacks

1. Keep up to date with the new updates on the operating system, third party software (MS Office, Browsers, and Browser plugins). Disable remote Desktop Connections.
2. Enable a personal firewall on workstations.
3. Maintain updated Antivirus software on all systems. 4. Strict External Device (USB drive) usage policy

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